HUAWEI

HUAWEI IRELAND TECH ARENA 2026

From user histories to recommendations

Generative sequential recommendation under constraints

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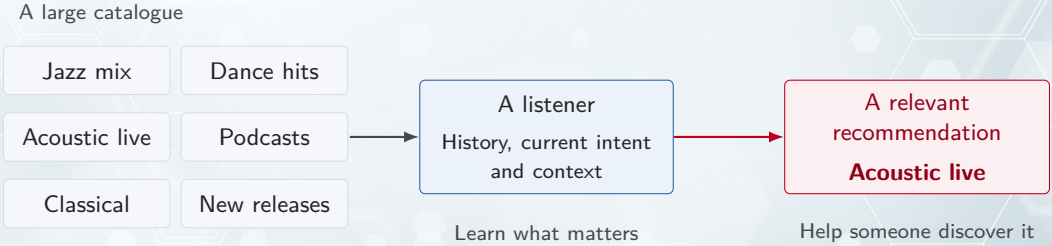
AIRecS, Huawei Ireland Research Centre

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Recommendation turns abundance into discovery

01 / WHY RECOMMENDATION MATTERS

Which item is useful to *this person*, *right now*?



Users

Find useful content, products and services.

Platforms

Make each interaction more relevant.

Creators and sellers

Reach people who may value their work.

The same problem appears in shopping, video, news and many other digital services.

A shared idea: predict what comes next

Language



A listener's interaction history



$$p_\theta(x_1, \ldots, x_T) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t})$$

x_t: a text token or an encoded interaction; $x_{<t}$: the preceding sequence.

Learn a conditional distribution from the history.

LLMs motivate a scalable recipe for recommendation

02 / WHY THIS DIRECTION IS EXCITING

Represent events
Items, actions and context

→

Learn from histories
Conditional prediction

→

Scale and measure
Data, context and model capacity

SASRec
2018

Attention over item histories for next-item prediction.

TIGER
2023

Generate structured item identifiers from interaction histories.

HSTU
2024

Scale sequence models adapted to recommendation data.

How far can the sequence modelling recipe take us in recommendation?

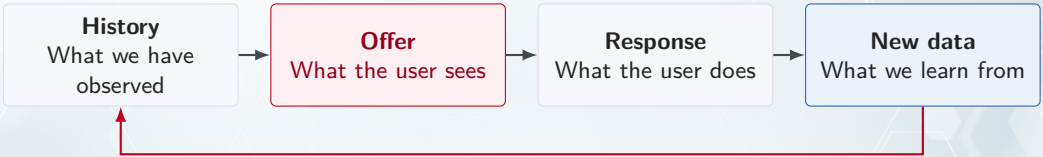
What changes when the sequence records behaviour?

	Typical text	Interaction histories
Vocabulary	A fixed vocabulary of reusable subword tokens	A changing catalogue of items, including new items
Time	Position within the text	Order, elapsed time, sessions
Meaning	Tokens in a linguistic context	Views, clicks, skips and purchases carry different signals
Input/Output Geometry	Short input sequences; long output sequences	Long input sequences; short output sequences

The representation must preserve which event happened, and when.

Recommendations help create tomorrow's training data

03 / OBSERVATION, FEEDBACK AND OBJECTIVES



- A logged click reflects both preference and exposure.
- For an item that was never shown, the response is unknown.
- Choosing recommendations changes the interactions we will observe.

Predicting a response and choosing an action are connected problems.

Schnabel et al., Recommendations as Treatments (ICML 2016).

Two phases: predict a response, then choose an offer

Phase 1 Predict clicks

How likely is this user to click this candidate?

Given
Interaction history and candidate items.

Build
A model of the user's response.

Submit
A file of predicted click probabilities.

Phase 2 Compete and adapt

Which item should your agent offer?

Given
A simulated marketplace and user histories.

Build
A recommendation agent competing for user choice.

Submit
Models for evaluation at round boundaries.

Use what you learn about user behaviour to make better decisions.

Phase 1: estimate the response to a candidate

Our listener's history

- Jazz mix: played
- Dance hits: skipped
- Acoustic set: replayed

$$p_\theta(\text{click} \mid H, i)$$

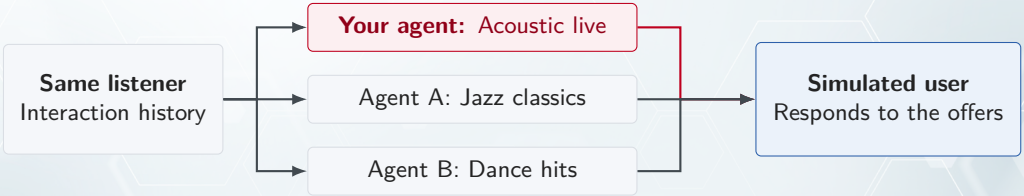
H: observed history i: candidate item

Illustrative predictions

Candidate	Click probability
Acoustic live	0.74
Jazz classics	0.50
Dance hits	0.12

Learn from labelled examples → Submit aligned predictions → Score against hidden labels

Phase 2: recommend, observe, improve



Four rounds of development and feedback



Start with a baseline. Improve quality and efficiency with each round.

Teams update their models between evaluation rounds.

Selected reading

REFERENCES / FOR PARTICIPANTS WHO WANT TO EXPLORE

- Kang and McAuley (2018).
  Self-Attentive Sequential Recommendation.
  SASRec: attention over interaction histories.
- Rajput et al. (2023).
  Recommender Systems with Generative Retrieval.
  TIGER: autoregressive generation of semantic item identifiers.
- Zhai et al. (2024).
  Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers for Generative Recommendations.
  HSTU: sequence modelling adapted to recommendation at scale.
- Schnabel et al. (ICML 2016).
  Recommendations as Treatments: Debiasing Learning and Evaluation.
  Selection bias and propensity-weighted learning and evaluation.